Geongyu Lee 이건규 · イ・ゴンギュ

Machine Learning Researcher · Computational Pathology × Multi-Omics

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I develop machine learning models that predict molecular measurements from pathology slides and, in my current industry work, drug response from proteomics. I focus on whether these predictions remain valid at new hospitals and for drugs not seen in training. I have 5+ years of industry R&D experience, and at Deep Bio I contributed to pre-deployment model validation for a medical-AI product submitted to Korea's Ministry of Food and Drug Safety (MFDS).

SELECTED PUBLICATIONS

Google Scholar · 62 citations · h-index 5 · i10-index 3 · as of Nov 2026

1. **Assessing the risk of recurrence in early-stage breast cancer through H&E stained whole slide images.** Scientific Reports 15, 35069, 2025. First author · doi.org/10.1038/s41598-025-16679-x
2. **MoSPR: Histology-to-Gene Expression Prediction with Morpho-Spatial Macrostates and Low-Rank Molecular Programs.** Preprint, 2026 · under review. Co-first author · arXiv:2609.34280 · code released
3. **Artificial intelligence–driven digital pathology in urological cancers: current trends and future directions.** Prostate International 13(4), 181–190, 2025. Review · co-first author · doi.org/10.1016/j.prnil.2025.02.002
4. **A reliability map for per-gene multiome RNA velocity parameters in single-cell kinetics.** ML4H 2026, Findings track. Accepted paper · poster presentation · co-author, presenter · also oral at BIOINFO/GIW ISCB-Asia 2026
5. **KPIs 2024 challenge: Advancing glomerular segmentation from patch- to slide-level.** Medical Image Analysis 114, 104234, 2026. Challenge report · co-author · team 2nd place, whole-slide track · doi.org/10.1016/j.media.2026.104234
6. **AI-guided deep visual proteomics reveals recurrence-associated morpho-molecular signatures within triple-negative breast cancers.** Preprint, 2026. Co-author · arXiv:2608.03145
7. **A Computationally Efficient Multi-Section Whole Slide Image Analysis for Capturing Prostate Cancer Multifocality in Biochemical Recurrence Prediction.** Preprint, 2026. Co-author · arXiv:2603.20273
8. **G2L: From Giga-Scale to Cancer-Specific Large-Scale Pathology Foundation Models via Knowledge Distillation.** AAAI 2026 Workshop (W3PHIAI). Accepted paper · oral (presented by a co-author) · co-author · arXiv:2510.11176
9. **MurSS: A Multi-Resolution Selective Segmentation Model for Breast Cancer.** Bioengineering 11(5), 463, 2024. Co-author · doi.org/10.3390/bioengineering11050463
10. **Supervised Contrastive Embedding for Medical Image Segmentation.** IEEE Access 9, 138403–138414, 2021. Co-author · doi.org/10.1109/ACCESS.2021.3118694

EXPERIENCE

OMIXAI (formerly RadiSen) — Feb 2025 – Present

AI Researcher · Seoul

- Lead R&D on a multi-omics model that combines proteomics and RNA with H&E pathology to predict drug response, in an 8-person team (4 proteomics, 4 AI; in progress).
- Built a proteomics-based model of cell-line drug response (IC50) and evaluated it leave-one-drug-out over 381 compounds, so test drugs were never seen in training (Pearson ≥ 0.65; internal R&D, unpublished). Also worked on self-supervised representation learning for proteomics with LoRA/PEFT.
- Developed ADMET prediction models and a drug-recommendation algorithm for canine cancer (top-k ≥ 70% on an in-house cohort; internal R&D, unpublished).
- Planned and ran OMIXAI's Arc Institute Virtual Cell Challenge entry (CRISPR-knockdown response in pluripotent stem cells).
- Co-first author of the MoSPR preprint on gene-expression prediction from H&E (code released); ran half of the experiments, incl. baselines. The method ranked 1st of 15 in the paper's benchmark on three TCGA cancers (gene-wise PCC).

Deep Bio — Mar 2021 – Jan 2025

AI Researcher · Seoul

- First-authored a two-hospital study (n=125) predicting Oncotype DX recurrence-score risk groups from H&E slides alone (Scientific Reports 2025; sensitivity 0.86 / 0.75 / 0.53 for low / intermediate / high risk). Built a WSI pipeline that processed 500+ slides.
- Developed a lymph-node metastasis detection model and, with the Deep Bio team, the glomerular segmentation model for the KPIs 2024 Challenge at MICCAI 2024 (2nd place, whole-slide track). Co-authored the challenge report (Medical Image Analysis, 2026).
- Led prostate metastasis and recurrence-risk modeling, incl. lymph-node metastasis detection (unpublished).

– Contributed to pre-deployment model validation and regulatory documentation for a medical-AI product submitted to Korea's MFDS.

Nuricon 2021

Intern · Pangyo

– Built a parking-lot fire-detection AI system.

SELECTED PROJECTS

Veterinary oncology CDSS: drug recommendation for canine cancer · OMIXAI · IPET national R&D · 2026–2030
Drug-recommendation algorithm for canine cancer toward a clinical decision-support system (top-k ≥ 70% on an in-house cohort; internal R&D, unpublished).

Brain-hemorrhage detection on CT · SeoulTech × SK / Ajou Univ. Hospital · 2019
Hemorrhage segmentation (U-Net, Dice 0.66, 50 test patients) and slice classification (accuracy 0.90, 143 patients). Ranking performance held on the external RSNA set; class-activation maps confirmed the model looked at the bleed.

EDUCATION

M.S. Data Science · Seoul National University of Science and Technology (SeoulTech) 2019 – 2021
Advisor: Prof. Sangheum Hwang · Thesis: "Utilizing Feature-wise Contrastive Loss for Improving Segmentation Performance."

B.S. Information Security · Daejeon University 2012 – 2019

SKILLS

Deep Learning	PyTorch, HuggingFace, PEFT / LoRA, Accelerate, DDP / FSDP
Pathology	OpenSlide, QuPath, CLAM / weakly-supervised MIL, UNI · CONCH · Virchow, WSI tiling
Bio / Omics	scanpy · AnnData, single-cell perturbation, proteomic representation learning, ADMET
MLOps & Languages	Python, R, SQL, Bash, Docker, Slurm, W&B, MLflow, FastAPI, Git

AWARDS & FUNDED RESEARCH

KPIs 2024 Challenge · Whole-Slide Track
2nd place · glomerular segmentation, held in conjunction with MICCAI 2024 · results published in Medical Image Analysis 2026

Pan-Sarcoma Proteogenomic Profiling for Precision Oncology
National Cancer Center Korea · Korea–US–Japan · human (Kyung Hee Univ. Med. Ctr.) + canine · 2025–2028 · Participating Researcher

Virtual-Cell CDSS for Veterinary Oncology
IPET / Ministry of Agriculture · 2026–2030 (ongoing) · Participating Researcher

General-Purpose AI for Cancer Pathology Diagnosis
Lead: Deep Bio · National R&D · program 2021–2025 · Participating Researcher, Apr 2021 – Jan 2025

CONFERENCE PRESENTATIONS

Predictability is not substitutability: a cost-of-substitution framework for H&E-based molecular prediction across 5 cancers
· BIOINFO/GIW ISCB-Asia 2026 · Accepted poster · Co-first & corresponding author · doi.org/10.5281/zenodo.24068874

Predicting Protein Receptor Status from H&E-stained Images in Breast Cancer · AACR Annual Meeting 2023 · Poster · First author

Recurrence Risk Prediction Based on Automatic Histologic Analysis of Breast Cancer Using Whole Slide Images
· AACR Annual Meeting 2022 · Poster · First author

A Deep Learning based Pancreatic Adenocarcinoma Survival Prediction Model Applicable to Adenocarcinoma of Other Organs
· AACR Annual Meeting 2022 · Poster · Co-author

Automatic Histological Grading of Breast Cancer Resection Tissue · USCAP Annual Meeting 2022 · Poster · First author

Breast Cancer Survival Analysis through the Extracted Feature from the Prostate Diagnosis Model
· USCAP Annual Meeting 2022 · Poster · Co-author

영역분할 모델 성능 향상을 위한 대조적 손실 함수의 활용 (Utilizing a contrastive loss to improve segmentation model performance)
· KIIE Fall Conference 2020 · Oral · First author

COMMUNITY & SERVICE

Research member, AutoBioX, Pseudo Lab Season 12 · 2026 · 2 studies at BIOINFO/GIW ISCB-Asia 2026, 1 also at ML4H 2026 Findings

Reviewer, ML4H 2026 (Machine Learning for Health) · 2026

AI Career & Project Mentor, Codeit · 2025 – Present · part-time · 30+ mentees 1:1, 8+ project teams across 3 cohorts

Co-author, two Korean e-books on AI drug discovery (BOOKK) · 2026 · wrote the AI and math parts